Criteria and self-assessment grid for NeuroML course: Final Project	Excellent	Good	Can be improved	Insufficient
Category				
Task setting	Perfectly understands and can formulate and justify the choice of the topic and the exact task. Knows state of the art on this problem, all needed steps to accomplish the task and approximate timeline. Can estimate the risks of the project	Understands the topic and the exact task of the project. Has studied other papers on the same or similar topics. Can estimate project demands and timeline.	Can work on the project, where the task was formulated by others (the team or advisors), understands task relevance	Does not understand the project task and its relevance
Data preprocessing	You suggest multiple options (including experimental ones) of preprocessing, including different steps and parameters. You can explain what and why you chose to these data, how it is tested and justified for optimal performance	You know main established preprocessing steps and defauls parameters for them. You can test different preprocessing pipelines and explain your choice	You know basic preprocessing methods and apply them to raw data	You take raw data w/o any needed preprocessing for further steps
Exploratory data analysis	Intense exploration and evidence of many trials and failures. You have looked at the data in many different ways before coming to your final answer. You have gone beyond what was asked: additional research from other sources used to help understand/explain findings.	Plenty of exploration and investigation. Some additional research helps explain findings, and some of your ideas are creatively presented and explained.	Some exploration, but little evidence that you have selected the best of many ideas. Little or no additional research.	There is no evidence of exploration. You have not come up with your own questions of the data
Reasoning about data	You suggest multiple explanations for a given finding, and use multiple tools to explore surprising results. You present one or two as the most plausible, but have allowed for the possibility that you are wrong. You are self-critical: What did I do well? What did I do poorly? What have I missed? How could I do better next time? You identify flaws in methodology and provide suggestions as to how they could be remedied. You don't blindly accept perceived wisdom, but challenge preconceived notions and come up with interesting new ways of testing them.	You are sceptical and self-critical, but not consistently. There is some critical analysis, and some use of multiple techniques to answer the same question.	Some findings accepted without question. Self- criticism weak.	Findings accepted uncritically. Leaps of logic without justification. You have not thought about how to do better next time.
Model training/evaluation	Clarifies the scope, available data, and values of the system. Selects a suitable model. Explains how the algorithm works and how to formulate the data. Defines how long the model will take to train and considers scaling opportunities. Suggests optimal metrics and justifies why they're useful for the given problem. May elaborate on useful ways to improve the model using the suggested metrics. Model training/evaluation is organized as a reproducible pipeline with weights and logs and can be easily reproduced.	Asks some thoughtful clarifying questions around the model's goals and available data. Suggests a basic model. Might have needed a hint, but otherwise would have gotten the answer on their own with more time. Understands the given problem and suggests a useful metric, possibly with some guidance. Model training/evaluation is organized as a reproducible pipeline with weights and logs and can be easily reproduced.	Suggests a model that may not work, but can speak to how adjustments can be made. Suggests a metric that won't work. Or suggests a metric, but may not understand how the dataset impacts the selection. Model training/evaluation is organized as a pipeline, can be hardly reproduced.	Fails to ask clarifying questions, jumps into solutions. Suggests a model that will not work, and clearly does not understand the different types of ML systems. Demonstrates zero experience evaluating models, unable to discuss relevant metrics. Model training/evaluation is not organized as a pipeline, can't be reproduced.
Model optimization	Constructs a loss function that can be optimized. Discusses appropriate regularization strategies, such as L1 regularization, that avoid overfitting or underfitting. Performs a research on hyperparameters selection and on model possible overfitting	Identify a loss function and reasonable optimization scheme for training. Clearly explains how the optimization scheme would work. Performs a research on hyperparameters selection.	Suggests a standard, basic technique (e.g. a loss function with regularization strategies that can't be easily optimized).	Demonstrates a clear lack of understanding about optimization techniques, such as loss functions, hyperparameters, and different optimizers.
Model interpretation	Can thoroughly explore the model in terms of its prediction: feature importance and post-hoc analysis, heatmap of class activation for neural networks. Statistical tests on features and data domain explaination.	Can select, justify and run model interpretation pipelines, shows a set of important data features, makes simple hypotheses on how the model works	Suggests a standard, basic technique for model interpretation	Does not perform model interpretation analysis
Programming	Introductory comment describes overall strategy and gives evidence of preliminary planning. Thoughtful problem decomposition breaks the problem into independent pieces that can be solved easily. Code free from duplication. Each function encapsulates a single task, and repeated tasks are performed by functions, not copy and paste. Notebook runs without issue. Code is in Git and can be easily reproduced.	Evidence of planning before coding, but some flaws in overall strategy. Code workable, but not elegant. Common programming idioms used to reduce code. Notebook runs without issue. Code is in Git, can be reproduced with medium efforts.	It all hangs together, but planning was absent or rushed. The code works, but copy-paste used often. Notebook runs without issue.	No evidence of planning. Strategy deeply flawed. Functions used inappropriately, or existing functions reinvented. Extensive use of copy and paste. Notebook does not run without issue.
Written Reports	Findings very well organised. Clear headings demarcate separate sections. Excellent flow from one section to the next. The paper is easy to scan. An abstract/summary/introduction at the start of the paper briefly summarises your approach, findings, and/or dataset. Conclusions at the end present further questions and ways to investigate more. Tables and graphics carefully tuned and placed for desired purpose.	Findings well organised and sections clearly separated, but flow is lacking. Each section has clear purpose. Tables and graphics clear and well chosen	Sections unclear and no attempt to flow from one topic to the next. Graphics and tables poorly chosen to support questions. Some have fundamental flaws.	It is hard to read your paper. There are no headings, figures are far away from where they are referenced in the text. There is no summary or conclusion.
Results visualization	Visualization makes appropriate reference to and significantly supports the information being presented.	Visualization makes appropriate reference to and generally support the information being presented.	Visualization occasionally refers to and partially support the information being presented.	The student's approach minimally refers to and support the information being presented.
Quality of presentation	Presentation slides are very clear, well crafted, organized, and compelling.	Presentation slides are mostly clear, reasonably well crafted and organised.	Presentation slides are unclear, lack polishing and organisation.	Presentation slides are confusing, clearly unpolished and disorganized.

Analysis of the results and future work	The speaker presents one or two of the most plausible explanations for a given finding, even if the projects aims were not achieved, but allows for the audience to make their own judgement. The methods employed are sound. The presenter is self-critical and honestly discusses flaws and limitations in methodology and provide suggestions as to how they could be remedied.	The speaker preciselyand honestly describes the results, even if the projects aims were not achieved. The methods are sometimes not fully adequate or consistently justified. The speaker is skeptical and self-critical, but not consistently.	The methods are questionable and not well justified. The speaker hasn't blinded accepted findings, but hasn't come up with many ways to check results either.	Trying to obscure true results when the aims were not achieved. The methods are inappropriate and not justified. Leaps of logic without justification. Findings accepted uncritically. Inappropriate methods. The speaker has not thought about how to do better next time